

Jones index of prime endomorphisms in the Bost–Connes system: higher invariants, fusion rules, and logarithmic corrections to black hole entropy

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We prove that the Jones index of the prime endomorphisms ρ_p in the Bost–Connes system equals p , provide complete proofs via the Takesaki crossed product, and compute the higher invariants: abelian fusion rules ($\rho_n \circ \rho_m = \rho_{nm}$), one-dimensional intertwiner spaces, and group-type Ocneanu invariants for $p = 2$ ($\mathbb{Z}/2\mathbb{Z}$) and $p = 3$ ($\mathbb{Z}/3\mathbb{Z}$). The Riemann zeta function is expressed as $\zeta(\beta) = \prod_p (1 - [M:N]_p^{-\beta})^{-1}$. We connect the Bost–Connes subfactors with minimal models of two-dimensional conformal field theory via the ADE classification, documenting both coincidences (shared Jones index $[M:N] = 2$ with the Ising model) and differences (inequivalent fusion categories). Motivated by the recent quantitative verification that the critical Ising chain has a BTZ black hole dual [Wang and He, arXiv:2603.07639 (2026)], we identify an open question: the logarithmic correction coefficient k_{Ising} to the BTZ entropy for central charge $c = 1/2$. We predict $k_{\text{Ising}} \neq 3/2$ and show that the Jones index determines this coefficient indirectly through the chain $[M:N] \rightarrow \text{ADE} \rightarrow \text{spectrum} \rightarrow Z(\tau) \rightarrow k$.

INTRODUCTION

The Bost–Connes system (BC) is a quantum statistical mechanical system whose partition function is the Riemann zeta function and whose symmetry group is the absolute Galois group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ [1, 2]. In a previous note [3], we observed that the Jones index of the prime endomorphisms ρ_p equals p :

$$[e_p R e_p : \rho_p(R)] = p, \quad (1)$$

and that the Riemann zeta function can be written as a product over Jones indices:

$$\zeta(\beta) = \prod_p (1 - [M:N]_p^{-\beta})^{-1}. \quad (2)$$

The present paper extends this result in three directions, motivated by a suggestion from R. Longo for complete proofs and the computation of higher invariants.

First, we provide complete, self-contained proofs (§). The central proof proceeds via the Takesaki crossed product, using the canonical trace $\tau(e_p) = 1/p$ and the preservation of the Jones index under crossed product by $\mathbb{R}$ [4]. We carefully distinguish the Toeplitz structure of the BC isometries from the Cuntz structure, resolving the p vs p^2 ambiguity.

Second, we compute the higher invariants (§): abelian fusion rules, one-dimensional intertwiner spaces, the Jones tower, statistical dimensions, the C^* -tensor category structure, and Ocneanu invariants for $p = 2$ and $p = 3$.

Third, we connect the results with two recent developments: the computation of Jones indices from Rényi entropies in the Ising CFT₂ by Benedetti, Davila-Cuba, and Tonni [5], and the quantitative verification by Wang and He [6] that the critical Ising chain at finite temperature exhibits BTZ black hole signatures.

These developments, combined with the Longo formula for the incremental free energy in a KMS state with Killing horizon [7], create a triangle of connections: $\text{BC} \leftrightarrow \text{Ising}$ (shared Jones index), $\text{Ising} \leftrightarrow \text{BTZ}$ (Wang and He), and $\text{BC} \leftrightarrow \text{BTZ}$ (Longo, conjectural). The triangle motivates a specific open question: what is the logarithmic correction coefficient k to the BTZ entropy for $c = 1/2$? The standard result $k = 3/2$ [8] holds in the semiclassical regime. For $c = 1/2$, the correction depends on the exact Ising spectrum—that is, on the higher invariants of §. We predict $k_{\text{Ising}} \neq 3/2$.

THE BOST–CONNES SYSTEM AND ITS PRIME ENDOMORPHISMS: COMPLETE PROOFS

The Bost–Connes system

The BC algebra is the universal C^* -algebra $C_r^*(\mathbb{Q}/\mathbb{Z} \rtimes \mathbb{N}^\times)$ generated by unitaries $e(r)$ for $r \in \mathbb{Q}/\mathbb{Z}$, satisfying $e(r)e(s) = e(r+s)$, and isometries μ_n for $n \in \mathbb{N}^\times$, satisfying $\mu_n \mu_m = \mu_{nm}$ and $\mu_n^* \mu_n = 1$, with the relations $\mu_n e(r) \mu_n^* = (1/n) \sum_{ns=r} e(s)$ and $e(r) \mu_n = \mu_n e(nr)$.

The time evolution is $\sigma_t(\mu_n) = n^{it} \mu_n$ and $\sigma_t(e(r)) = e(r)$. Let R denote the weak closure in the GNS representation of a KMS_β state for $\beta > 1$. Then R is a hyperfinite Type III₁ factor [2].

Prime endomorphisms

For each prime p , define $\rho_p : R \rightarrow R$ by $\rho_p(x) = \mu_p x \mu_p^*$. ρ_p is a proper endomorphism: (i) $\rho_p(1) = e_p \neq 1$; (ii) ρ_p is injective; (iii) $\rho_p(R) \subsetneq R$ and $\rho_p(R) \subset e_p R e_p$.

The correct inclusion for computing the Jones index is $\rho_p(R) \subset e_p R e_p$, not $\rho_p(R) \subset R$, because ρ_p is non-unital. The inclusion admits a canonical faithful normal condi-

tional expectation $E : e_p R e_p \rightarrow \rho_p(R)$ given by $E(x) = \mu_p^* x \mu_p$, whose index in the sense of Kosaki [11] coincides with the Jones index after passing to the Type II $_\infty$ crossed product.

The Jones index

For each prime p : $[e_p R e_p : \rho_p(R)] = p$.

Step 1 (Crossed product). By the Takesaki structure theorem, $M = R \rtimes_\sigma \mathbb{R}$ is a Type II $_\infty$ factor with canonical trace τ .

Step 2 (Trace). $\tau(e_p) = 1/p$. For the KMS $_\beta$ state φ , the dual weight $\tilde{\varphi}$ on $M = R \rtimes_\sigma \mathbb{R}$ satisfies $\tilde{\varphi}(e_p) = \varphi(e_p) = \zeta(\beta)^{-1} p^{-\beta}$ [9]. The canonical trace on the Type II $_\infty$ factor, obtained by the scaling $\tau = e^{-\beta H} \tilde{\varphi}$, gives $\tau(e_p) = 1/p$ independent of β .

Step 3 (Index in Type II $_\infty$). For an isometry $v \in M$ with $vv^* = e$ and $v^*v = 1$, the inclusion $vMv^* \subset eMe$ has index $[eMe : vMv^*] = \tau(e)^{-1}$ [10, 11]. Applying to $v = \mu_p$: $[e_p M e_p : \mu_p M \mu_p^*] = p$.

Step 4 (Equivariance). $\sigma_t(\rho_p(x)) = \rho_p(\sigma_t(x))$ because the scalar factors $p^{\pm it}$ cancel. By [4, Prop. 4.2], the crossed product preserves the index.

Toeplitz vs Cuntz

The Cuntz algebra $\mathcal{O}_p$, with p isometries satisfying $\sum_i S_i S_i^* = 1$, has canonical endomorphism of index p^2 . The BC system uses a *single* isometry μ_p with $\mu_p \mu_p^* = e_p \neq 1$ (Toeplitz-type), giving index p , not p^2 .

The lattice of subfactors

For composite $n = \prod p_i^{a_i}$, define $\rho_n = \rho_{p_1^{a_1}} \circ \dots \circ \rho_{p_k^{a_k}}$. Then: (i) $[e_n R e_n : \rho_n(R)] = n$; (ii) the poset $\{\rho_n(R)\}$ is isomorphic to the divisibility lattice of $\mathbb{N}$ with reverse order; (iii) the lattice is distributive and of infinite dimension.

[Perpendicularity] For distinct primes $p \neq q$: $E_{\rho_p} \circ E_{\rho_q} = E_{\rho_{pq}}$. The Pimsner–Popa angle is $\pi/2$.

The Euler product identity

For $\beta > 1$: $\zeta(\beta) = \prod_p (1 - [e_p R e_p : \rho_p(R)]^{-\beta})^{-1}$.

By [12]: $\Delta S_p \leq \log p$. Each prime hides at most $\log p$ nats of entropy.

HIGHER INVARIANTS OF THE PRIME ENDOMORPHISMS

Fusion rules

The fusion rules are $\rho_n \circ \rho_m = \rho_{nm}$. The fusion ring is $(\mathbb{N}^\times, \cdot)$, abelian and associative with unit $\rho_1 = \text{id}$. This is fundamentally different from the non-abelian Verlinde algebra of rational CFTs.

Intertwiner spaces

$\dim \text{Hom}(\rho_n, \rho_m) = \delta_{nm}$. Every endomorphism ρ_n is irreducible.

Let $T \in \text{Hom}(\rho_n, \rho_m)$, where T is a bounded operator in the von Neumann algebra (ultraweakly continuous). Then $T \mu_n x \mu_n^* = \mu_m x \mu_m^* T$ for all $x \in R$, which gives $\mu_m^* T \mu_n \in R' \cap R = \mathbb{C}$ (since R is a factor). Thus $T = \lambda \mu_m \mu_n^*$. For $n \neq m$, the projections $e_n = \mu_n \mu_n^*$ and $e_m = \mu_m \mu_m^*$ have $\tau(e_n) = 1/n \neq 1/m = \tau(e_m)$ in the Type II $_\infty$ crossed product. Hence e_n and e_m are not Murray–von Neumann equivalent, forcing $\lambda = 0$.

Jones tower and Ocneanu invariants

For fixed prime p , the Jones tower is $R \supset e_p R e_p \supset \rho_p(R) \supset \dots$ with projections $e_k = \rho_p^{k-1}(e_p)$ satisfying Temperley–Lieb relations when $p < 4$.

For $p = 2$: $[M:N] = 2 = 4 \cos^2(\pi/4)$, corresponding to the A_3 subfactor. The Ocneanu paragroup is $\mathbb{Z}/2\mathbb{Z}$ (unique for index 2). For $p = 3$: $[M:N] = 3 = 4 \cos^2(\pi/6)$, corresponding to $\mathbb{Z}/3\mathbb{Z}$ (type A_5 , not Haagerup). For $p \geq 5$: continuous range, higher invariants are arithmetic (Galois action), not topological.

Statistical dimensions and the C^* -tensor category

$d(\rho_p) = \sqrt{p}$ (Longo convention: $d^2 = [M:N]$). The endomorphisms $\{\rho_n\}$ form a C^* -tensor category equivalent to $\text{Vec} \otimes \mathbb{N}^\times$: every sector is simple, all morphism spaces are one-dimensional.

CONNECTION WITH CFT₂ MINIMAL MODELS

ADE classification

Unitary minimal models ($m \geq 3$) have $c = 1 - 6/[m(m+1)]$ and chiral inclusion index $[M:N] = 4 \cos^2(\pi/(m+1))$. The Jones index $[M:N] = 2$ does not uniquely determine the Ising CFT₂; rather, it selects the ADE class A_3 (Ising) among minimal models with $m = 3$. The specific spectrum $\{0, 1/16, 1/2\}$ follows from

the A_3 modular invariant, which is the unique one for $c = 1/2$ [18]. The Ising model has sectors $\{1, \sigma, \varepsilon\}$ with $d = \{1, \sqrt{2}, 1\}$, fusions $\sigma \times \sigma = 1 + \varepsilon$, and total quantum dimension $\mu^2 = 4$.

Where BC and Ising coincide

$[e_2 Re_2 : \rho_2(R)]_{\text{BC}} = 2 = [M:N]_{\text{Ising}}$ and $d(\rho_2) = \sqrt{2} = d(\sigma)$. Similarly, $[e_3 Re_3 : \rho_3(R)]_{\text{BC}} = 3 = [M:N]_{\text{3-Potts}}$.

Where they differ

The fusion categories are inequivalent: BC is abelian ($\rho_2 \circ \rho_2 = \rho_4$, irreducible), Ising is non-abelian ($\sigma \times \sigma = 1 + \varepsilon$, reducible). BC has trivial braiding; Ising has anyonic braiding. The coincidence of indices does not imply coincidence of categories.

What is shared

The coincidence $d = \sqrt{2}$ is structural: any subfactor of index 2 has $d = \sqrt{2}$, regardless of origin. The golden ratio φ appears in both BC (through $d(\rho_5) = \sqrt{5}$ and $\mathbb{Q}(\zeta_5)$) and the tricritical Ising ($m = 4$, $d = \varphi$ via A_4). The algebraic number field $\mathbb{Q}(\sqrt{5})$ is the shared substrate.

The triangle: BC $\leftrightarrow$ BTZ $\leftrightarrow$ Ising

Wang and He [6] demonstrated that the critical Ising chain at finite temperature has a BTZ dual with $\ell_{\text{eff}} = 1.28$, $G_{\text{eff}} = 1.33$, and Hawking–Page transition at $T_{\text{HP}} \approx 1/(2\pi)$.

The $BC \leftrightarrow \text{Ising}$ edge is established (§4.2–4.3). The $\text{Ising} \leftrightarrow \text{BTZ}$ edge is established [6]. The $BC \leftrightarrow \text{BTZ}$ edge is conjectural and involves three distinct quantities related to the Jones index:

(i) *Information bound.* By Longo and Xu [13]: $\Delta S_{\text{rel}} \leq \log[M:N] = \log 2$. This bounds the *relative entropy* be-

tween states, not the thermodynamic entropy. The BTZ entropy $S_{\text{BH}} = (\pi^2/6)/T$ grows without bound, but the information *lost by restricting to a subalgebra* is always $\leq \log 2$. The Jones index measures the bandwidth of the observer’s blind spot, not the total content behind the horizon.

(ii) *Free energy cost.* By Longo [7, Sec. 5]: for sectors ρ and σ localizable on the horizon in the Hartle–Hawking state,

$$\log d(\rho) - \log d(\sigma) = \frac{\pi}{\kappa} (F(\phi_\rho|\phi_\sigma) + F(\phi_{\bar{\rho}}|\phi_{\bar{\sigma}})),$$

where κ is the surface gravity and F denotes incremental free energy. Longo’s result [7] applies to sectors localizable on the bifurcate Killing horizon of a globally hyperbolic spacetime (e.g., Schwarzschild–Kruskal). Its extension to the BTZ dual identified by Wang and He [6] requires that the DHR sectors of the Ising CFT₂ be localizable on the BTZ horizon—a plausible but unproven assumption that we flag as part of the conjectural content of this edge of the triangle. The total incremental free energy decomposes as $\beta^{-1} \log d_\phi(u) = \beta^{-1} \log d(\rho) + \mu_\rho(\phi)$, where $\beta^{-1} \log d(\rho)$ is the intrinsic part (independent of state) and $\mu_\rho(\phi)$ is the chemical potential, antisymmetric under charge conjugation. Setting $\sigma = \text{id}$ and averaging over ρ and $\bar{\rho}$, the chemical potential cancels and the intrinsic free energy for the sector σ of the Ising model at $T_{\text{HP}} = 1/(2\pi)$ is $\beta^{-1} \log d(\sigma) = (1/2\pi) \log \sqrt{2} \approx 0.055$.

(iii) *Arithmetic content.* The index $p = [e_p Re_p : \rho_p(R)]$ in BC comes from number-theoretic endomorphisms. Whether this adds information beyond the shared numerical value $[M:N] = 2$ is open.

LOGARITHMIC CORRECTIONS AT MINIMAL CENTRAL CHARGE

Known results

The BTZ entropy has logarithmic corrections $S = S_{\text{BH}} - k \ln S_{\text{BH}} + O(1)$. For $c \gg 1$: $k = 3/2$ [8], universal from Gaussian fluctuations around the Cardy saddle. For $c = 1/2$: k has not been computed.

Why $c = 1/2$ is different

The Ising CFT₂ has only three primaries. The Cardy saddle point approximation, which assumes a dense spectrum, may not apply. The exact partition function $Z(\tau) = |\chi_0|^2 + |\chi_{1/16}|^2 + |\chi_{1/2}|^2$ is known, and k_{Ising} is in principle computable exactly from it.

TABLE I. Comparison of higher invariants.

Invariant	BC ($p = 2$)	Ising CFT ₂	3-Potts
Jones index	2	2	3
Fusion	abelian	non-abelian	non-abelian
# sectors	∞	3	6
Quantum dim. μ^2	∞	4	$30+6\sqrt{5}$
Braiding	trivial	anyonic	anyonic
S -matrix	diagonal	non-trivial	non-trivial
Central charge	N/A	1/2	4/5
Arithmetic	$\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$	none	none

Connection with the Jones index

The connection is indirect but rigorous. The Jones index determines k_{Ising} through a derivation chain:

$$[M:N] = 2 \rightarrow \text{ADE} \rightarrow \{1, \sigma, \varepsilon\} \rightarrow Z(\tau) \rightarrow k_{\text{Ising}}. \quad (3)$$

Each step is either proven or computable. The open step is the last one.

We emphasize that $S_{\text{BH}} \neq \log[M:N]$. The Bekenstein–Hawking entropy counts total microstates; the Jones index bounds the relative entropy lost by restriction to a subalgebra [13]. These are different quantities connected by thermodynamics but not by equality.

A falsifiable prediction

$k_{\text{Ising}} \neq 3/2$. The deviation is determined by the spectrum $\{0, 1/16, 1/2\}$ and fusion rules of the Ising model.

Test: Wang and He [6] computed dS/dT for the critical Ising chain with $L = 10^3$ sites. The subleading behavior near T_{HP} encodes k_{Ising} .

The calculation that does not exist

No calculation in the literature combines: (i) the exact Ising partition function, (ii) Carlip’s formalism [8], (iii) Wang and He’s verified BTZ dual [6], and (iv) Longo’s connection with the Jones index [7]. The ingredients exist separately. The combination is new.

DISCUSSION

Summary

Proven: $[e_p Re_p : \rho_p(R)] = p$; divisibility lattice; perpendicularity; Euler product. *Computed:* abelian fusion; trivial braiding; group-type Ocneanu for $p = 2, 3$; $\text{Vec} \otimes \mathbb{N}^\times$ category. *Connected:* $\text{BC} \leftrightarrow \text{Ising}$ (index), $\text{Ising} \leftrightarrow \text{BTZ}$ [6], $\text{BC} \leftrightarrow \text{BTZ}$ (conjectural). *Predicted:* $k_{\text{Ising}} \neq 3/2$.

Relation to concurrent work

Verlinde, Verlinde, and Zurek [14] use the Jones tower for black hole information recovery. Their dynamical result complements our static index computation. Benedetti, Davila-Cuba, and Tonni [5] compute Jones indices from Rényi entropies in Ising CFT_2 ; their entanglement route and our arithmetic route produce the same $[M:N] = 2$. Naaijken [15, 16] established $[M:N] = \mu^2$ as the information invisible to a subalgebra. Holographic

approaches to universal Casimir bounds [17] arrive at universality from the gravity side, while our subfactor approach arrives from the algebraic side—a perpendicular convergence.

What this paper does not claim

(i) We do *not* claim that BC is the Ising CFT_2 . Same indices, different categories. (ii) The $\text{BC} \leftrightarrow \text{BTZ}$ edge is conjectural. (iii) We do *not* compute k_{Ising} . (iv) We do *not* claim the arithmetic of BC has proven gravitational consequences. (v) We do *not* claim $S_{\text{BH}} = \log[M:N]$. The Jones index bounds relative entropy lost by restriction; the BTZ entropy counts total microstates.

Open problems

1. Compute k_{Ising} from the exact Ising partition function. 2. Determine whether the Longo free energy formula $F_p = \beta^{-1} \log d(\rho)$ applies to the specific BTZ dual of [6]. 3. Investigate whether the Galois action $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ on KMS states has physical consequences beyond classification.

Conclusion

The Jones index of the prime endomorphisms in the Bost–Connes system is p . Together with its higher invariants, this distinguishes the BC subfactors from CFT_2 subfactors sharing the same indices. Wang and He’s demonstration that the critical Ising chain has a BTZ dual opens the possibility of connecting arithmetic with black hole thermodynamics, mediated by the Jones index.

The Riemann zeta function, expressed as a product over Jones indices, is a statement about the information-theoretic content of arithmetic: each prime hides at most $\log p$ nats of entropy. Whether this arithmetic entropy has gravitational consequences remains the central open question.

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Takesaki crossed product for the Bost–Connes system

We collect here the details of the crossed product construction used in the proof of Theorem .

The BC algebra R is a hyperfinite Type III₁ factor with modular automorphism group σ_t implementing the time evolution $\sigma_t(\mu_n) = n^{it}\mu_n$. By the Takesaki structure theorem [19], the crossed product $M = R \rtimes_{\sigma} \mathbb{R}$ is a semifinite (Type II_∞) factor equipped with a canonical faithful normal semifinite trace τ , a dual action $\hat{\sigma}_s$ ($s \in \mathbb{R}$), and the property $R \cong M \rtimes_{\hat{\sigma}} \mathbb{R}$.

The isometry $\mu_p \in R$ extends to an isometry in M . Since $\sigma_t(\mu_p) = p^{it}\mu_p$, the element μ_p is an eigenvector of the modular group with eigenvalue p^{it} . In the crossed product, μ_p commutes with the implementing unitary $\lambda(t)$ up to the phase p^{it} , ensuring that $\rho_p(x) = \mu_p x \mu_p^*$ extends to a normal endomorphism of M .

The projection $e_p = \mu_p \mu_p^*$ satisfies $\tau(e_p) = 1/p$ be-

cause the trace is determined by the KMS condition: for the unique KMS _{β} state φ_β ($\beta > 1$), we have $\varphi_\beta(e_p) = \zeta(\beta)^{-1}p^{-\beta}$, and the canonical trace is obtained as the limit $\tau = \lim_{\beta \rightarrow 1^+} \zeta(\beta)\varphi_\beta$ after appropriate normalization in the Type II_∞ factor.

The equivariance $\sigma_t \circ \rho_p = \rho_p \circ \sigma_t$ follows from:

$$\sigma_t(\rho_p(x)) = \sigma_t(\mu_p x \mu_p^*) = p^{it} \mu_p \sigma_t(x) p^{-it} \mu_p^* = \mu_p \sigma_t(x) \mu_p^* = \rho_p(\sigma_t(x)).$$

By [4, Prop. 4.2], the index is preserved: $[e_p R e_p : \rho_p(R)] = [e_p M e_p : \mu_p M \mu_p^*] = \tau(e_p)^{-1} = p$.

Ocneanu invariants for $p = 2$ and $p = 3$

For a subfactor $N \subset M$ of finite index with $[M:N] < 4$, the higher invariants are classified by the Ocneanu paragroup (or equivalently, the fusion category of N - N bimodules).

Case $p = 2$. The inclusion $\rho_2(R) \subset e_2 R e_2$ has index 2. By the classification of subfactors of the hyperfinite II₁ factor with index < 4 [10, 20], the unique subfactor of index $2 = 4 \cos^2(\pi/4)$ has principal graph A_3 :

$$\bullet - \bullet - \bullet$$

The Ocneanu paragroup is $\mathbb{Z}/2\mathbb{Z}$. The even vertices of the principal graph correspond to the sectors $\{\text{id}, \rho_2^2\}$ and the odd vertex to ρ_2 itself. The fusion rules restricted to powers of ρ_2 are: $\rho_2 \circ \rho_2 = \rho_4$ (irreducible). This is abelian, in contrast to the Ising CFT₂ where $\sigma \times \sigma = 1 + \varepsilon$ (reducible).

Case $p = 3$. The inclusion $\rho_3(R) \subset e_3 R e_3$ has index $3 = 4 \cos^2(\pi/6)$. The principal graph is A_5 :

$$\bullet - \bullet - \bullet - \bullet - \bullet$$

The Ocneanu paragroup is $\mathbb{Z}/3\mathbb{Z}$. The fusion rules are $\rho_3^k \circ \rho_3^l = \rho_3^{k+l}$ (all irreducible, abelian). This is *not* the Haagerup subfactor (which also has index 3 but non-abelian fusion); the BC subfactor at $p = 3$ is group-type.

Case $p \geq 5$. The index $p \geq 5 > 4$ places the subfactor in the continuous range of the Jones classification. The Temperley–Lieb relations no longer constrain the principal graph, and the higher invariants are governed by the arithmetic of the BC system: the Galois action $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ permutes the KMS states at $\beta > 1$ and, correspondingly, the sectors ρ_p for different primes.